

DevOps Lab Program

Program8: Create kubernetes Pods using Yaml Descriptors

Structure of Program

Program8/

→ nodejs-configmap.yaml

→ nodejs-pod.yaml

→ server.js

Steps for execution:

Step 1: Create a server.js file



```
GNU nano 8.1
const http = require("http");
const port = 3000;

const server = http.createServer((req, res) => {
  res.end("Hello from Node.js running inside Kubernetes!");
});

server.listen(port, () => {
  console.log(`Server running at port ${port}`);
});
```

Step 2: Build the config map file

```

GNU nano 8.1
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Hello from Node.js running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`Server running at port ${port}`);
    });

```

Step 3: Build the pod.yaml file

```

GNU nano 8.1
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-pod
  labels:
    app: nodejs
spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node", "/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000
  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config

```

Step 4: Apply yaml files

Kubectl apply -f nodejs-configmap.yaml

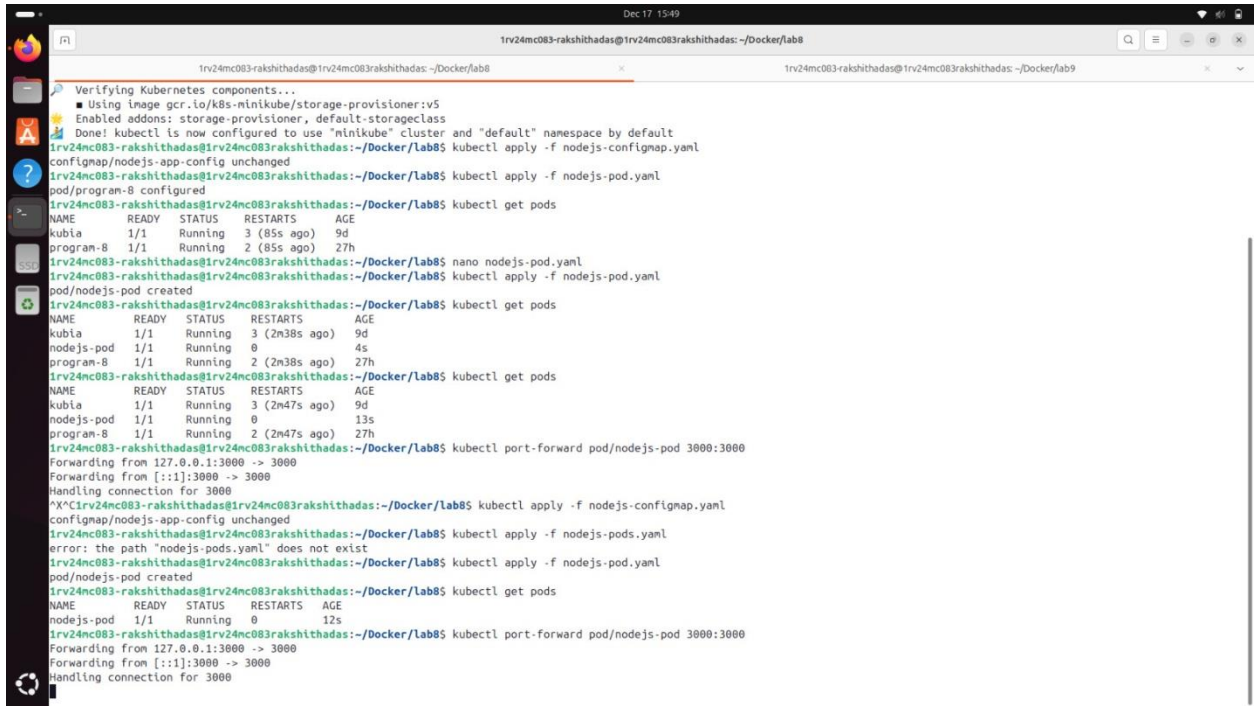
Kubectl apply -f nodejs-pod.yaml

Step 5: Check status

Kubectl get pods

Kubectl get deployments

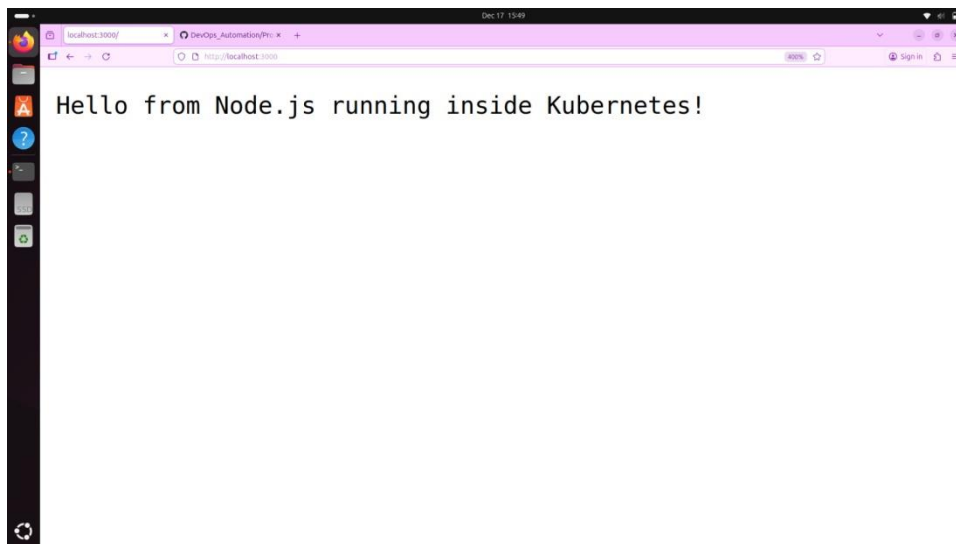
Step 6: Access the web page



```
1rv24mc083-rakshithadas@1rv24mc083rakshithadas: ~/Docker/lab8
Verifying Kubernetes components...
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ■ Enabled addons: storage-provisioner, default-storageclass
  ■ Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
kubia     1/1     Running   3 (85s ago)  9d
program-8 1/1     Running   2 (85s ago)  27h
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ nano nodejs-pod.yaml
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
kubia     1/1     Running   3 (2m38s ago)  9d
nodejs-pod 1/1     Running   0          4s
program-8 1/1     Running   2 (2m38s ago)  27h
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
kubia     1/1     Running   3 (2m47s ago)  9d
nodejs-pod 1/1     Running   0          13s
program-8 1/1     Running   2 (2m47s ago)  27h
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::]:3000 -> 3000
Handling connection for 3000
^X^C1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config unchanged
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl apply -f nodejs-pods.yaml
error: the path "nodejs-pods.yaml" does not exist
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl apply -f nodejs-pod.yaml
pod/nodejs-pod created
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
nodejs-pod 1/1     Running   0          12s
1rv24mc083-rakshithadas@1rv24mc083rakshithadas:~/Docker/lab8$ kubectl port-forward pod/nodejs-pod 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::]:3000 -> 3000
Handling connection for 3000
```

Step 7: Check the output in:

<http://localhost:3000>



Step 8: Delete all the pods

Step 9: Delete all the services

Step 10: Delete all the deployments